Summer Internship Program of Institute of Statistical Science Final Personal Report

From Variants to Structure: Assessing Protein Structures of Hypertension-Associated Genes Using AlphaFold3 and TM-score/RMSD

student：RUAN,BO-CHENG

Adviser：Hsin-Chou Yang PhD Chun-Houh Chen PhD

August 2025

Table of content

1 Introduction **1**

2 Materiel and Method **2**

2.1 Data Sources 2

2.2 Data Description 2

2.3 Data Preprocessing 3

2.4 Data Analysis Methods 4

2.5 Computational Environment and Tools 6

3 Result **7**

3.1 Consistency of AlphaFold3 predictions under fixed vs random seeds 7

3.2 Global and local structural comparisons 8

4 Discussion **10**

5 Conclusion **11**

List of Figures

1.1 Age-specific prevalence of hypertension in Taiwan (2017—2020). 1

2.1 Workflow of 3D protein modeling and structural comparison for hypertension associated variant genes 3

2.2 Top 10 hypertension-associated genes in the dataset. 5

3.1 Comparison of AlphaFold3 predictions under fixed vs. random seeds across three target genes. 7

3.2 NPR1 Comparison of global and domain-level RMSD distributions between participants with and without hypertension. 8

3.3 NPR1 Comparison of global and domain-level TM-score distributions between participants with and without hypertension. 8

3.4 RGL3 Comparison of global and domain-level RMSD distributions between participants with and without hypertension. 9

3.5 RGL3 Comparison of global and domain-level TM-score distributions between participants with and without hypertension. 9

3.6 ADRB1 Comparison of global and domain-level RMSD distributions between participants with and without hypertension. 9

3.7 ADRB1 Comparison of global and domain-level TM-score distributions between participants with and without hypertension. 10

List of Table

2.1 Main fields of the TWB VCF file (GRCh38/hg38) 3
2.2 Target genes, transcripts, and their association with hypertension 5

1 Introduction

In Taiwan, hypertension is a major public health issue. According to the Nutrition and Health Survey in Taiwan (NAHSIT, 2017—2020), the prevalence of hypertension among adults aged 18 years and older is 26.8%, corresponding to an estimated 5.29 million individuals. The prevalence increases with age, and deaths due to hypertension related cardiovascular diseases, including heart disease, cerebrovascular disease, and hypertensive disease, have long ranked among the top ten causes of mortality. These findings highlight the importance of preventing and better understanding hypertension. As shown in Figure 1.1, the prevalence of hypertension rises sharply with age and is consistently higher in men than in women.

Genome wide association studies (GWAS) have identified numerous loci associated with disease, but most findings are associative and do not directly explain the mechanisms that cause disease. Biologically, genes determine a protein ' s sequence; the sequence shapes its structure; and structure drives function. Therefore, to probe disease mechanisms, we should examine how variants alter protein structure. The recent advent of AlphaFold 3 has made it feasible to convert variant sequences into 3D protein structures for analysis.

In this study, we map participants ' genetic variants onto 3D protein structures and evaluate them via structural alignment. We built an end-to-end pipeline to derive hypertension-related variant protein sequences: candidate genes associated with hypertension were curated from the GWAS Catalog and integrated with VCF variant data from 1,491 Taiwan Biobank (TWB) participants. After phasing, we generated haplotype-specific (H1/H2) variant protein sequences for each transcript of each candidate gene. Using AlphaFold 3, we modeled 3D structures for the reference and variant sequences, and performed global and local Cα alignments using RMSD and TM-score to examine whether structural differences correlate with participants ' hypertension status. We currently use three target genes, *RGL3*, *ADRB1*, and *NPR1*, to demonstrate feasibility and future scalability.

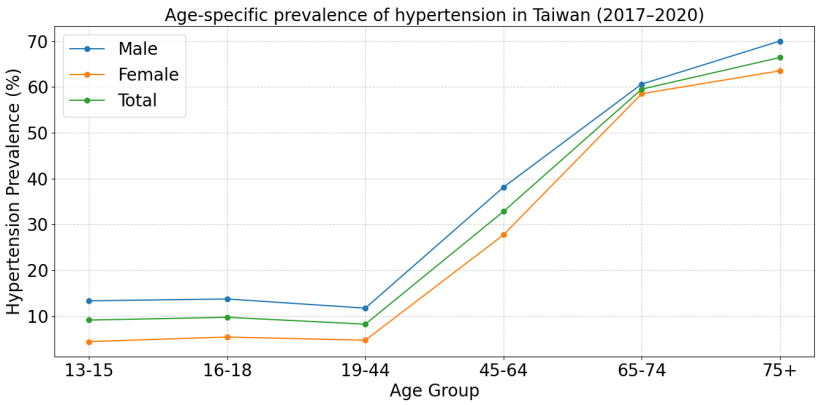


Figure 1.1: Age-specific prevalence of hypertension in Taiwan (2017—2020).

2 Materiel and Method

2.1 Data Sources

This study used three main data sources. First, the GWAS Catalog is a public database jointly maintained by the National Human Genome Research Institute (NHGRI) and the European Bioinformatics Institute (EMBL-EBI). Since its establishment in 2008, it has continuously collected and organized published genome-wide association studies (GWAS), providing standardized information on variant—trait associations as well as study samples and methodological descriptions. It is widely used for identifying disease related genes and for downstream functional studies(Buniello et al., 2019). In this study, hypertension-related candidate genes were obtained from this database. Second, Ensembl is a genome resource jointly maintained by EMBL and EBI. It provides genome assemblies and annotations for multiple species, including gene coordinates, transcript structures, variation data, and comparative genomics resources(Cunningham et al., 2022). We used Ensembl to obtain the chromosome positions and start end coordinates of the candidate genes. Finally, the Taiwan Biobank (TWB) is a large-scale prospective cohort study that has collected genomic and diverse phenotypic data from more than 150,000 participants, primarily of Han Chinese ancestry. TWB data can also be linked to national health registries to support long term epidemiological and genomic research(Feng et al., 2022). In this study, we used whole genome variant data (VCF) provided by TWB and combined it with questionnaire and physical examination data from 1,491 participants. The resulting hypertension-related variant VCF data were used for downstream analyses.

2.2 Data Description

The genomic variant data provided by the Taiwan Biobank (TWB) are in Variant Call Format (VCF, version 4.2, reference genome GRCh38/hg38). A VCF file consists of two parts, one is the header, which records metadata such as file format and filter criteria, and the other is variant records, which store the variant information in tab delimited fields. Example header entries include:

- `FILTER`: <ID=PASS, Description="All filters passed">, indicating that the variant passed all quality filters.
- `ALT`: <ID=NON_REF, Description="Represents any possible alternative allele at this location">, denoting potential alternative alleles at this position.

Variant records contain the main fields listed in Table 2.1.

Table 2.1: Main fields of the TWB VCF file (GRCh38/hg38)

Field	Description
CHROM	Chromosome identifier
START	Start position of the variant
END	End position of the variant
ID	Variant identifier (e.g., dbSNP rsID)
REF	Reference allele
ALT	Alternate allele(s)
QUAL	Quality score of the variant call
FILTER	Filter status (e.g., PASS)
INFO	Additional annotations (e.g., allele frequency, functional effects)
FORMAT	Defines the content of genotype fields (e.g., GT, GQ)
Sample columns	One column per participant, storing genotypes (e.g., 0/0, 0/1, 1/1, or 0|1 after phasing)

In this study, we mainly used the `CHROM`, `START`, `END`, `REF`, `ALT`, and genotype (`GT`) fields to extract variants within candidate hypertension related genes.

2.3 Data Preprocessing

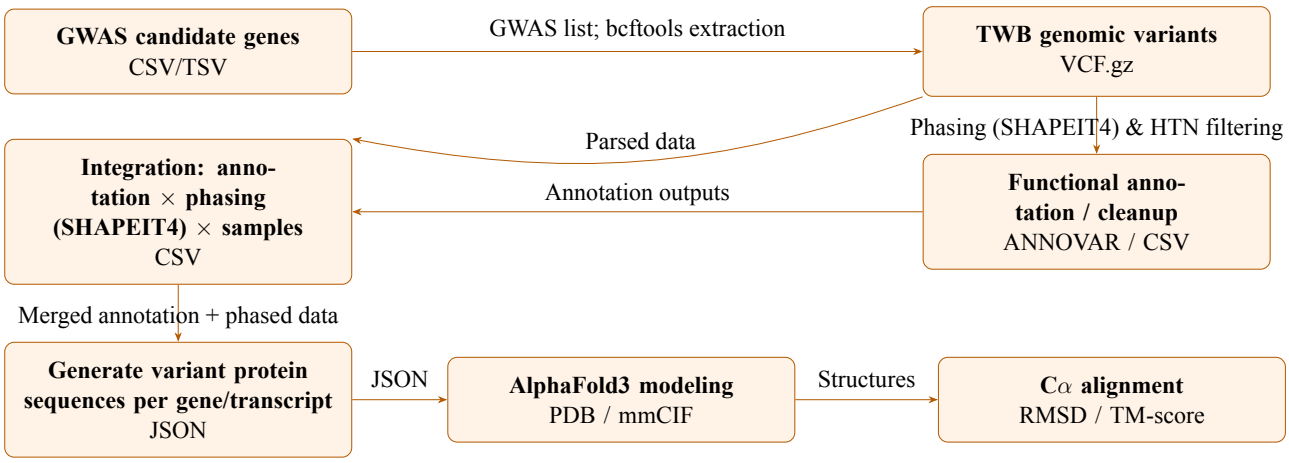


Figure 2.1: Workflow of 3D protein modeling and structural comparison for hypertension associated variant genes

We were mainly interested in whether structural differences are related to hypertension, so we first needed to obtain variant protein sequences. Therefore, our approach was as follows.

First, we obtained candidate genes associated with hypertension from the GWAS Catalog. We then used Ensembl to retrieve the chromosome, start, and end positions of these genes and converted the results into BED format. Next, we applied bcftools to extract variants from the Taiwan Biobank (TWB) VCF files that were located within these gene regions, producing a VCF file containing variants in hypertension related genes. We then used SHAPEIT4 to phase the genotype (GT) information in the sample columns, reconstructing haplotype sequences (H1 and H2) for each gene on the two chromosome copies. Although the parental origin cannot be determined, this step prevents variants from the two alleles from being mixed. The phased VCF file was used as the main dataset for downstream analysis.

Next, we used ANNOVAR to annotate the phased VCF. Annotation converts genomic coordinates into functional information at the gene and protein level, such as the affected gene, transcript, exon, cDNA change, and protein change, to evaluate the potential impact on structure and function. We focused on the `AAChange.refGene` field, which contains these details, and split it into separate columns for further analysis. Since ANNOVAR output does not keep the sample columns, we merged the annotation results with the phased dataset and saved the combined file as CSV format.

After annotation, we mapped each gene to its reference transcript sequence and applied amino acid substitutions according to the phased haplotypes of each participant. This produced $N \times 2$ protein sequences, where N is the number of samples, corresponding to the two haplotypes per individual. We merged these sequences with the CSV file and also generated a JSON file containing the variant protein sequences. These files were then used as input for AlphaFold3 (serve version) to generate 3D protein models, which were further analyzed by structural comparison.

The overall workflow is shown in Figure 2.1, starting from hypertension-associated genes identified by the GWAS Catalog, through variant extraction, phasing, annotation, and protein sequence generation, to 3D modeling with AlphaFold3 and structure comparison.

2.4 Data Analysis Methods

Because AlphaFold3 has daily usage limits, we could not model all candidate genes within the available time. Therefore, we selected representative genes based on the following criteria. First, only genes associated with hypertension in the GWAS Catalog were considered. Second, we filtered for significant associations with $p < 5 \times 10^{-8}$. Third, we focused on missense variants, since they

are more likely to alter protein structure. After applying these filters, about 100 genes remained. The top ten most frequent genes are shown in Figure 2.2, accounting for about 40% of the total. Finally, we selected three genes, RGL3, ADRB1, and NPR1, for downstream analysis, and their details are summarized in Table 2.2.

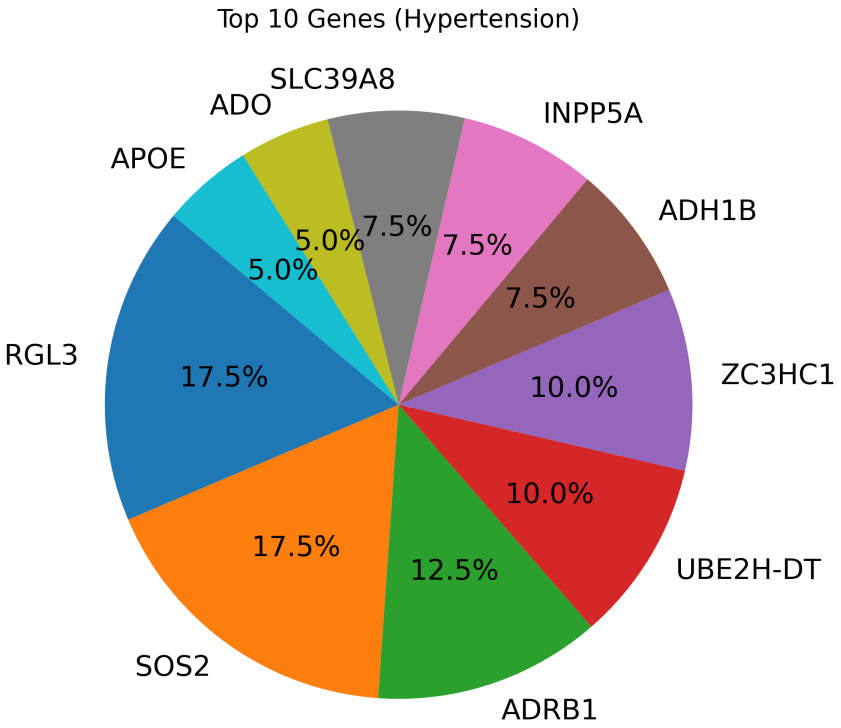


Figure 2.2: Top 10 hypertension-associated genes in the dataset.

Table 2.2: Target genes, transcripts, and their association with hypertension

Gene	**Transcript**	**Chromosome**	**Length (aa)**	**Association with Hypertension**
RGL3	NM_001035223	chr19	710	Implicated in BP regulation based on GWAS findings.
ADRB1	NM_000684	chr10	477	Beta-1 adrenergic receptor; variants influence BP and cardiac output.
NPR1	NM_000906	chr1	1061	Natriuretic peptide receptor 1; regulates sodium excretion and BP.

We also examined whether fixing the random seed in AlphaFold3 affects structural outputs by generating 20 fixed seed models and 380 random seed models, followed by pairwise comparisons using t-tests.

After confirming the stability of model predictions with fixed and random seeds, we further examined whether structural differences exist between original and variant protein models. We first performed global Cα alignments to evaluate overall structural similarity, and then conducted local (domain-level) comparisons to capture regional differences. Two metrics were used to quantify structural similarity: root mean square deviation (RMSD) and the template modeling score (TM-score). Their formulas are given below.

$$\text{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|p_i - q_i\|^2} \tag{1}$$

$$\text{TM-score} = \frac{1}{L_{\text{target}}}\sum_{i=1}^{L_{\text{align}}}\frac{1}{1+\left(\frac{d_i}{d_0(L_{\text{target}})}\right)^2} \tag{2}$$

where

- N: number of aligned atoms
- p_i, q_i: coordinates of atom i after optimal rotation/translation
- $d_i = \|x'_i - y'_i\|$: distance between atom i in aligned and reference structure
- $d_0(L) = \max\big(0.5, 1.24(L-15)^{1/3} - 1.8\big)$: length-dependent scale
- L_{target}: length of the target structure (residues)
- L_{align}: number of residues actually aligned

2.5 Computational Environment and Tools

All analyses were conducted using a combination of programming languages and bioinformatics tools.

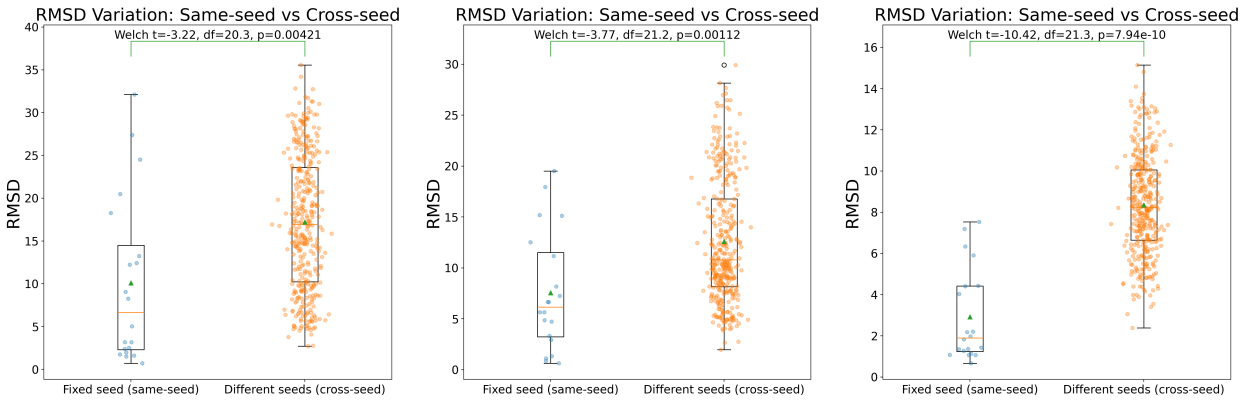
Python was employed for data processing and visualization, with major packages including `pandas`, `matplotlib`, and `Biopython` . Shell scripting was used to integrate workflows and handle large scale genomic data.

Several bioinformatics tools were applied at different steps of the analysis:

- **bcftools (v1.19)**: Processing and filtering VCF files, and extracting candidate gene regions.
- **SHAPEIT4 (v4.2.0)**: Genotype phasing, converting unphased genotypes (0/1) into phased haplotypes (0|1), and generating H1/H2 sequences.
- **ANNOVAR (2020Jun)**: Functional annotation of genomic variants with gene- and protein-level effects.
- **AlphaFold3 (2024)**: Prediction of 3D protein structures for variant sequences using deep learning modeling.
- **PyMOL (v3.1.0, Open-Source, 2025-05-02)**: Protein structure superposition and alignment; calculation of RMSD and TM-score.

3 Result

3.1 Consistency of AlphaFold3 predictions under fixed vs random seeds



(a) **NPR1**
Fixed seed vs. Random seed
($p = 0.00421$)

(b) **RGL3**
Fixed seed vs. Random seed
($p = 0.00112$)

(c) **ADRB1**
Fixed seed vs. Random seed
($p = 7.94 \times 10^{-10}$)

Figure 3.1: Comparison of AlphaFold3 predictions under fixed vs. random seeds across three target genes.

Based on t-tests of the three target genes, all p-values were less than 0.05. This shows that AlphaFold3 produces significantly different results when using fixed or random seeds(Figure 3.1). Therefore, in the following structural comparisons, we always used a fixed seed to ensure stable and reproducible results.

3.2 Global and local structural comparisons

First, we looked at the RMSD and TM-score boxplots for NPR1. In the global alignment results, RMSD showed only small differences and there was no clear relation to hypertension status. TM-scores were close to 1, which means the overall structures were almost the same and did not differ between the hypertension and non-hypertension groups. In the domain-level results, both RMSD and TM-score suggested that the cytoplasmic region of NPR1 may have some structural changes. However, these changes did not show a clear pattern related to hypertension.

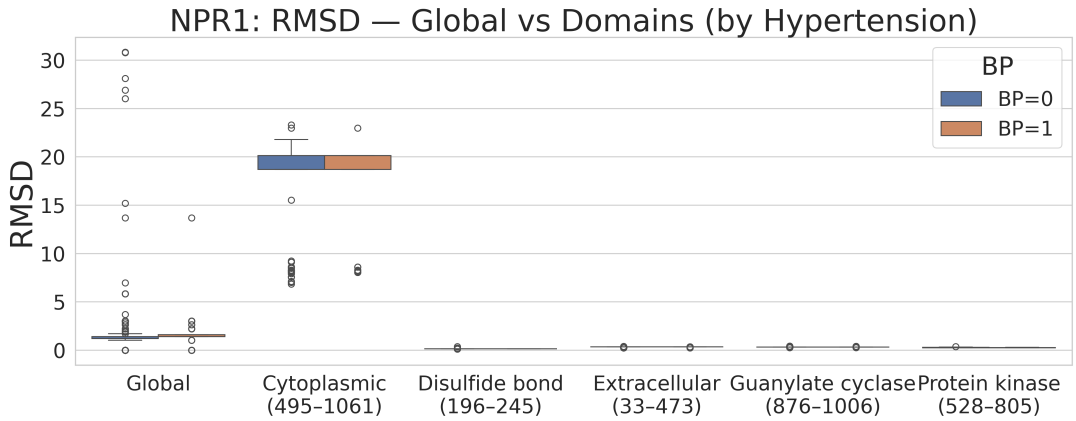


Figure 3.2: NPR1 Comparison of global and domain-level RMSD distributions between participants with and without hypertension.

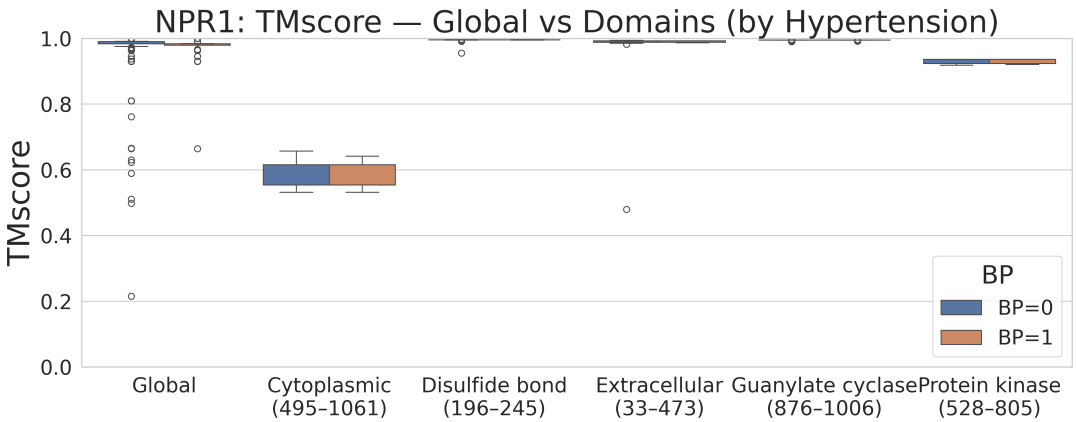


Figure 3.3: NPR1 Comparison of global and domain-level TM-score distributions between participants with and without hypertension.

Next, for RGL3, the global RMSD showed small differences but no relation to hypertension. TM-scores were also close to 1, showing that the structures were almost the same with no group differences. In the domain-level results, neither RMSD nor TM-score showed any structural changes, and there was no link to hypertension.

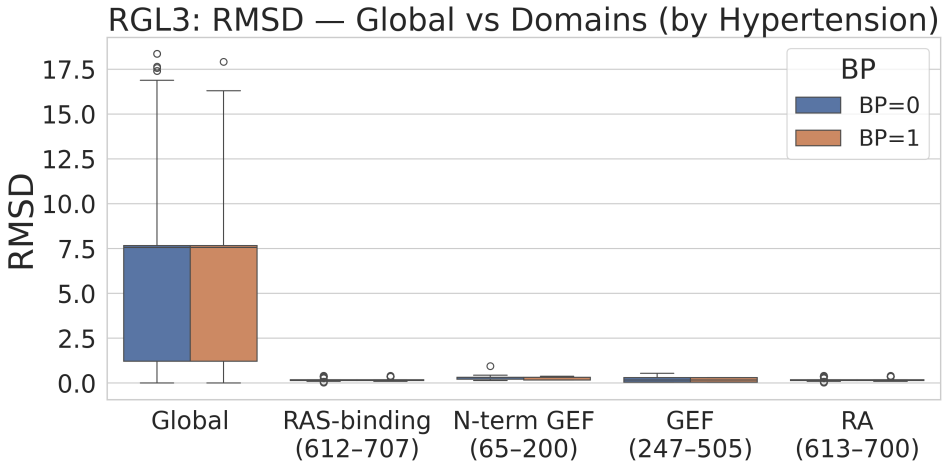


Figure 3.4: RGL3 Comparison of global and domain-level RMSD distributions between participants with and without hypertension.

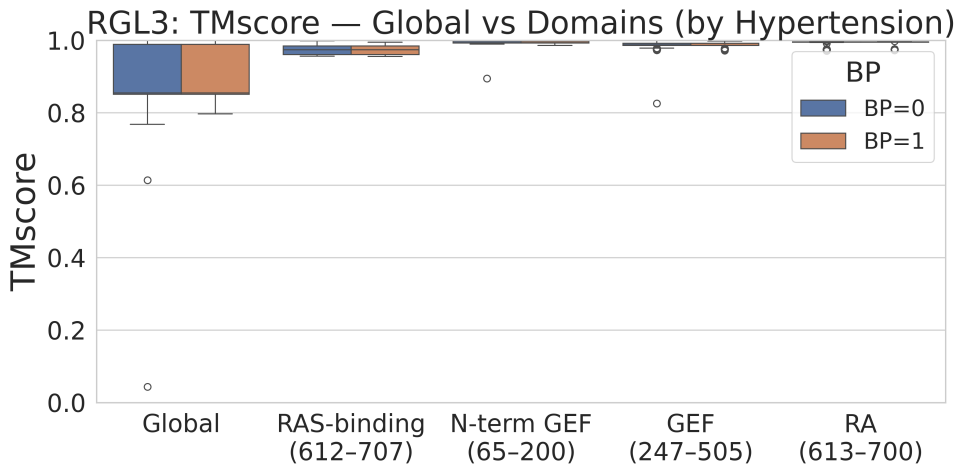


Figure 3.5: RGL3 Comparison of global and domain-level TM-score distributions between participants with and without hypertension.

Finally, for ADRB1, both RMSD and TM-score showed no structural differences in either global or domain-level results, and there was no relation to hypertension.

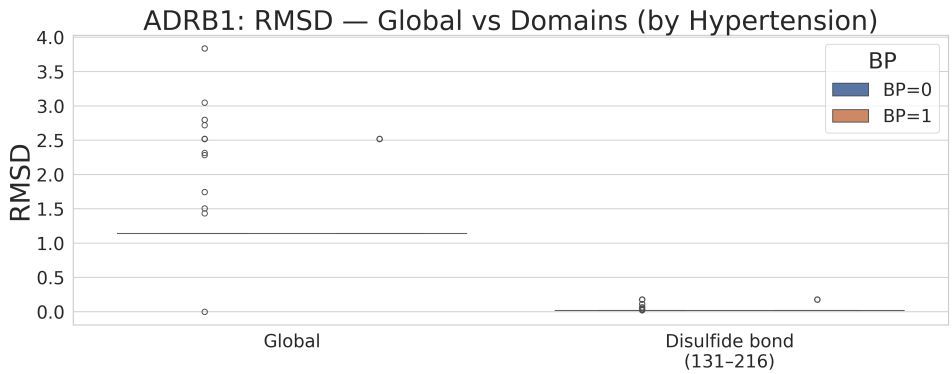


Figure 3.6: ADRB1 Comparison of global and domain-level RMSD distributions between participants with and without hypertension.

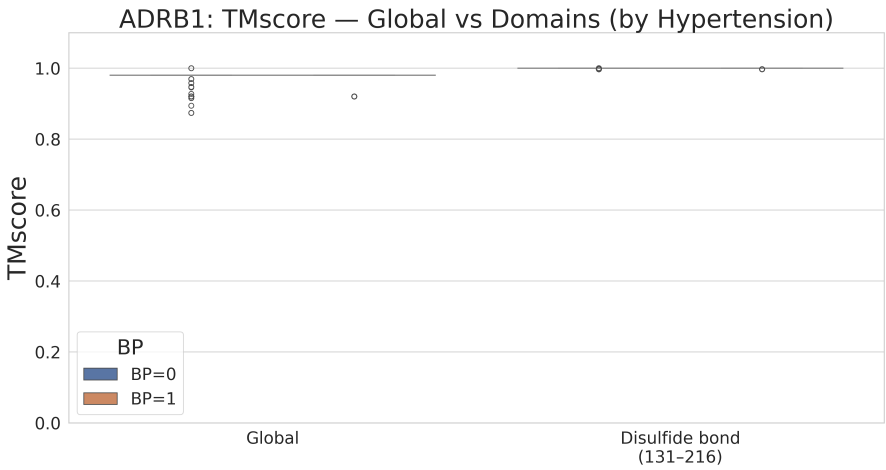


Figure 3.7: ADRB1 Comparison of global and domain-level TM-score distributions between participants with and without hypertension.

Overall, no clear structural differences were found between the hypertensive and non-hypertensive groups for the three target genes. Local differences were only seen in the cytoplasmic region of NPR1, but these were not clearly related to hypertension.

4 Discussion

First, as shown in Figure 3.1, although the t-test indicates significant differences between fixed and random seeds, this mainly reflects differences in mean distributions and does not mean that one setting is “better.” In practice, the purpose of fixing the seed is not to compare superiority but to reduce random variability, preventing AlphaFold3 modeling from being influenced by stochastic factors that could confound the comparison between reference and variant structures, and to ensure reproducibility of the results. Future studies may further explore whether seed-related differences extend to biological interpretations and investigate other approaches to address the limitations of using a fixed seed.

In the structural comparison, we first performed global alignments, and the results showed no clear differences between hypertensive and non-hypertensive groups for the three genes. We then conducted domain-level alignments, but again no significant differences were observed. We think this may be due to the characteristics of the selected genes, or that in this dataset these genes are not the main causes of hypertension.

This study has two main limitations. First, we only analyzed three target genes, which does not cover all candidate genes related to hypertension. Second, due to limited sample size and time, we could not test more genes or perform repeated model constructions.

Future research could include a larger set of genes and a bigger sample size to provide a more

comprehensive analysis of structural differences in hypertension-related genes. In addition, combining experimental structural data, such as cryo-EM or X-ray crystallography, would help validate AlphaFold3 predictions and strengthen the conclusions of this study.

5 Conclusion

At the beginning, we described the data sources and the structure of the VCF files, and introduced the reproducible end-to-end workflow established in this study. The workflow starts from GWAS genes and VCF files from the Taiwan Biobank (TWB), reconstructs H1/H2 haplotype variant protein sequences, and outputs JSON files that can be uploaded to the AlphaFold3 serve platform for structural modeling. We also explained the criteria for gene selection, focusing on missense variants that are more likely to affect protein structure. Due to time constraints, three target genes, RGL3, ADRB1, and NPR1, were selected as case studies for analysis.

The results showed that fixing the random seed produced more stable and reproducible models. In the structural comparisons, RGL3 and NPR1 displayed some differences, but these were not sufficient to clearly separate hypertensive and non-hypertensive groups. At the domain level, only the cytoplasmic region of NPR1 showed structural shifts, but these changes were not significantly associated with hypertension. Future work could expand the analysis to more genes and larger sample sizes to improve statistical power, and incorporate additional structural alignment methods or functional prediction metrics to complement RMSD and TM-score.

In summary, this study presents a feasible and reproducible workflow for structural comparison, demonstrating potential value for exploring protein structural differences.

References

[1] Danecek, P., Bonfield, J. K., Liddle, J., *et al.* (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. doi:10.1093/gigascience/giab008

[2] Delaneau, O., Zagury, J.-F., Robinson, M. R., Marchini, J., *et al.* (2019). Accurate, scalable and integrative haplotype estimation (SHAPEIT4). *Nature Communications*, 10, 5436. doi:10.1038/s41467-019-13225-y

[3] Wang, K., Li, M., & Hakonarson, H. (2010). ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. *Nucleic Acids Research*, 38(16), e164. doi:10.1093/nar/gkq603

[4] Abramson, J., AlQuraishi, M., Baker, D., *et al.* (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630, 493–500. doi:10.1038/s41586-024-07487-w

[5] DeLano, W. L. (2002). The PyMOL Molecular Graphics System. DeLano Scientific. http://www.pymol.org

[6] Kabsch, W. (1976). A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5), 922–923.

[7] Zhang, Y., & Skolnick, J. (2004). Scoring function for automated assessment of protein structure template quality. *Proteins*, 57(4), 702–710. doi:10.1002/prot.20264

[8] Health Promotion Administration, Ministry of Health and Welfare. (2021). *Nutrition and Health Survey in Taiwan (NAHSIT), 2017–2020: Prevalence of hypertension among adults*. Taipei: HPA, MOHW. Retrieved from `https://www.hpa.gov.tw/EngPages/Detail.aspx?nodeid=3999&pid=15562`